

Math 443 Homework 2

Mark Stanley

February 14, 2025

Problem 1

Let A be an $m \times n$ matrix of rank r . Suppose there exist right-hand sides b for which $Ax = b$ has no solution.

a

What are all inequalities ($<$ or $\leq$) that must be true between m , n , and r ?

Proof. Initially we note that b is of shape $(m, 1)$. We know that being in reduced row echelon form does not affect the solution to the system of linear equations, so envision now the augmented matrix $A|b$. The lack of solutions implies that at least 1 row from A is all 0, while there is some value for b , in other words we would have some $0x_1 + 0x_2 + \cdots + 0x_m = b \implies 0 = b$, where b is assumed to be nonzero (if it is not then the system has solutions!). Through our discussion, we immediately know that $r \leq m$ and $r \leq n$. But, from the fact that there is at least some 0 row, we also know that the rank must be strictly less than m . Now, let us consider 3 options.

- 1) $m = n$, it is immediate from assumption that therefore $r < n$ on top of $r < n$
- 2) $m > n$. We yield no extra information from this.
- 3) $m < n$. Again, we yield no extra information from this.

Last noteworthy point, these relationships are not identity claims, that is, there are examples of $r < m$ in which A does have valid x solutions (namely those where the 0th row relates to some $b_i = 0$) □

a

How do you know that $A^T y = 0$ has solutions other than $y = 0$

Proof. Again, through discussion we know that $\text{Rank}(A) = \text{Rank}(A^T)$. Thus, we can consider the rref of the same shapes 1) $n < m$, $r = n$. Because the rank of this transposition is full, it is the case that there is only 1 valid $y = 0$. 2) $n < m$ $r < n$. This case will yield multiple valid solutions, as there will necessarily be multiple indeterminants in a single column, $(a_1x_1 + \cdots + a_nx_n = 0)$, which has infinite solutions for $x_1, \dots, x_n$. 3) $m \geq n$, again we immediately know that $r < n$ in this case, and thus we have a row full of zeros, and thus infinite solutions for the 0 □

Problem 2

Find the complete solution (also called the general solution) to the system:

$$\begin{bmatrix} 2 & 6 & 4 & 8 \\ 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

Proof. We can use the compute the rref form of the matrix

□

Python Code

```
1 def rre(A_orig: np.array) -> Optional[np.array]:
2     """
3     Performs row reduction to get a matrix into reduced row echelon form.
4     Pseudocode from https://rosettacode.org/wiki/Reduced\_row\_echelon\_form
5
6     :param A_orig: the original square matrix A
7     :return: the rre form of matrix A
8     """
9     A = A_orig.copy()
10    lead = 0
11    rowCount = len(A)
12    columnCount = len(A[0])
13
14    for r in range(rowCount):
15        if lead >= columnCount:
16            return A
17
18        i = r
19        while A[i][lead] == 0:
20            i += 1
21            if i == rowCount:
22                i = r
23                lead += 1
24                if columnCount == lead:
25                    return A
26        A[[i, r]] = A[[r, i]]
27
28        if A[r][lead] != 0:
29            A[r] = A[r] / A[r][lead]
30
31        for i in range(rowCount):
32            if i != r:
33                A[i] = A[i] - A[i][lead] * A[r]
34        lead += 1
35
36    return A
```

Python Code

```
1 def rank(A: np.array) -> int:
2     """
3     Computes the rank of a matrix A. Does so by using row reduction to
4     get the reduced matrix, and then counting the number of non-zero rows.
5
6     :param A: the matrix A
7     :return: the rank of A
8     """
9     A_rre = rre(A)
10    return len([row for row in A_rre if any(row)])
```

This ran on our matrix results in an augmented matrix

$$A|b_{rref} = \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

Which yields no solution. How unfortunate.

Problem 3

Consider the matrix

$$A = \begin{bmatrix} 2 & -2 & 4 & 0 \\ 1 & -1 & 2 & 0 \\ 3 & -3 & 7 & 0 \end{bmatrix}$$

a

Find a basis for the column space $C(A)$

Proof. Let c_1, c_2, c_3, c_4 , be the columns of the matrix. We can immediately and easily tell that $\text{Span}\{c_1, c_2, c_3, c_4\} = \text{Span}\{c_1, c_2, c_3\}$, thus, we can ignore it for further computation. We can use the rre code to see that the determinant of this set is 0, meaning it is still linearly dependent. It is also immediately clear that because $c_1 = -c_2$, we can through one out, say c_2 . The rref of these two vectors clearly shows that $r = 2$, and therefore we know that this system is linearly independent. Thus, we know the vectors $c_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ $c_2 = \begin{bmatrix} 4 \\ 2 \\ 7 \end{bmatrix}$ is linearly independent, and spans $C(A)$

□

b

Find a basis for the null space $N(A)$

Proof. To solve for the null space, we rre on $A|0$, which yields

$$A|b_{rref} = \left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

It is clear that x_4 and x_3 need to be 0 in any case. Which just leaves $x_1 - x_2 = 0$, which yields infinite solutions. If we fix x_2 , we just have $x_1 = -x_2$ or $x = \begin{pmatrix} -x_2 \\ x_2 \\ 0 \\ 0 \end{pmatrix}$ Thus, we are

justified in saying the following is a basis $\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, as any vector that we can find as a solution must be of that form. □

c

Find a basis for the row space $C(A^T)$

Proof. Let c_1, c_2, c_3 , be the columns of the transposed matrix. We can perform the same process as for question 1. We can easily see that $c_1 = -c_2$, and we can throw one out as they are therefore linearly dependent, say c_2 , thus we are left with c_1 and c_2 , which are linearly dependent. □

d

Write the complete solution to $Ax = b$, where

$$b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Proof. Unfortunately there are none. Because the rref of the matrix has rank 2, the only valid solutions occur when the processes of rre on the augmented matrix also make the respective solutions 0, which did not happen with these numbers. Thus, the system holds no valid answers. □

Problem 4

Suppose A is the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix}$$

a

(16 points) Find all special solutions to $Ax = 0$ and describe in words the whole null space of A .

Proof. Assume $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$ Using our same rref function, we get our $A|0_{rref}$ to be

$$A|0_{rref} = \left[\begin{array}{cccc|c} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

This yields constraining equations $x_3 + 0.5x_4 = 0$, $x_2 + x_4 = 0$, and x_1 is a free variable. Thus, setting x_4 as a free variable yields $x_3 = -0.5x_4$ and $x_2 = -x_4$, which gives us a final vector

format of $\begin{pmatrix} x_1 \\ -x_4 \\ -0.5x_4 \\ x_4 \end{pmatrix}$, with two free variables x_1, x_4 □

b

(10 points) Describe the column space of this particular matrix A . "All combinations of the four columns" is not a sufficient answer.

Proof. Let the columns of A be denoted as c_1, c_2, c_3, c_4 . Using our rank algorithm $\text{Rank}(A) = 2$. Thus, we can immediately also make the claim that $\dim(\text{col}(A)) = 2$. This is pretty clear, as c_1 clearly adds no dimensionality, and $2c_2 + c_3 = 2c_4$, thus we can say c_2, c_3 are linearly independent and the $\text{span}(\text{col}(A)) = \text{span}(\text{col}(c_2, c_3))$ □

c

(10 points) What is the reduced row echelon form of $R = \text{rref}(B)$ when B is the 6×8 block matrix

$$B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$$

using the same A ?

Proof. The code provided with the rank shows that $\text{Rank}(A) = 2$. This rref code ends up with a $B = \begin{bmatrix} A_{rre} & A_{rre} \\ 0 & 0 \end{bmatrix}$. This makes intuitive sense. I won't prove it, but a rough explanation is that clearly the bottom rows A do not add any dimensionality along the rows, as they are the exact same as the ones above. So clearly the rank will not increase, and clearly not decrease. □

Python Code

```
1 if __name__ == '__main__':
2     # Problem 4
3     A = np.array([[0, 1, 2, 2], [0, 3, 8, 7], [0, 0, 4, 2]], dtype=float)
4     b = np.array([[0], [0], [0]], dtype=float)
5     augmented_matrix = np.concatenate((A, b), axis=1)
6     augmented_matrix_rre = rre(augmented_matrix)
7     print(augmented_matrix_rre)
8     print(rank(A))
9
10    # part c
11    B_row = np.concatenate((A, A), axis=1)
12    B = np.concatenate((B_row, B_row), axis=0)
13    print(B.shape)
14    print(B)
15
16    print(rre(B))
17    print(rank(B))
```